

ReadyCare Functional Specification (MVP)

1. Introduction

1.1 Purpose

ReadyCare is a web-based care marketplace platform that connects clients seeking care support with verified care professionals. The platform enables professionals to register, upload verification documents, manage availability, and accept bookings, while clients can search for professionals, make bookings, manage appointments, and leave reviews.

This functional specification defines the system behaviour, user roles, business rules, and functional requirements for the Minimum Viable Product (MVP).

2. Scope

The MVP supports:

- User registration and profile management
- Professional document verification
- Availability management
- Professional search
- Booking creation and management
- Mock payment handling
- Review and ratings
- Administrative verification workflows
- Real time notifications via email

Excluded from MVP:

- Live payment provider integration
- Messaging/chat
- Mobile application
- Dynamic pricing
- Advanced filtering
- Video consultations

3. User Roles

3.1 Client

A client can:

- Register an account
- Manage their profile and address

- Search for available professionals
- Create bookings
- Cancel bookings
- View booking history
- Mark a booking as no-show
- Leave reviews for professionals after a booking

3.2 Professional

A professional can:

- Register an account
- Upload required verification documents
- Upload optional qualification documents
- Manage their profile
- Set and manage availability
- Accept or reject bookings
- Mark bookings as completed
- View booking history
- Leave reviews for clients after a booking

A professional cannot appear in search or receive bookings until their required documents have been approved.

3.3 Admin

An admin can:

- Review uploaded professional documents
- Approve or reject professional documents
- Oversee professional verification status
- View users and booking-related information
- Issue refunds on bookings
- Delete user accounts (both professionals and clients)
- Create other admin accounts

4. Functional Requirements

4.1 Registration and Accounts

FR-1: Create Client Account

The system shall allow a client to register with:

- First name
- Last name
- Date of birth
- Gender
- Email
- Phone number
- Password
- Address

The system shall validate required fields, ensure the email address is unique, and create the related user, client profile, and address records.

FR-2: Create Professional Account

The system shall allow a professional to register with:

- Personal details
- Profile bio
- Years of experience
- Hourly rates
- Address

The system shall create a user record, professional profile record, and address record. The professional profile shall initially have an overall verification status of `NOT_SUBMITTED`.

FR-3: Create Admin Account

The system shall support admin accounts for users who need to review and approve professional documentation.

4.2 Professional Verification

FR-4: Required Documents

The system shall require professionals to provide the required document types before they can be approved.

The required document types are stored as reference data in the `document_types` table. For the MVP, the required documents are:

- Passport
- DBS certificate
- Care certificate

These document types shall have `is_required = true`.

FR-5: Optional Documents

The system shall allow professionals to upload optional documents or qualifications, such as:

- First aid certificate
- Manual handling training

- Safeguarding adults training

These document types shall have `is_required = false`.

FR-6: Document Upload

The system shall allow a professional to upload documents against a selected document type.

The actual document file shall be stored in Azure Blob Storage. The database shall only store the file storage reference and related metadata.

Each uploaded document shall store:

- Professional reference
- Document type reference
- File storage key
- Verification status
- Expiry date, where relevant
- Review information
- Rejection reason, where relevant

FR-7: Admin Document Review

The system shall allow an admin to review uploaded documents and either approve or reject them.

If rejected, the admin shall be able to provide a rejection reason.

FR-8: Professional Verification Rule

A professional shall only be considered approved when all active required document types have a corresponding uploaded document that is approved and not expired.

The professional profile field `overall_verification_status` shall be updated based on the status of the required documents.

4.3 Availability Management

FR-9: Create Availability Slots

The system shall allow a professional to create availability slots.

Each availability slot represents one hour. Slot start times must be on the hour.

Example valid slots:

- 09:00 to 10:00
- 10:00 to 11:00

Example invalid slot:

- 09:15 to 10:15

FR-10: Update Availability

The system shall allow a professional to manage their availability by adding, blocking, or removing slots.

Booked slots must not be available to other clients.

4.4 Professional Search

FR-11: Search Professionals

The system shall allow clients to search for available professionals by:

- City
- Date
- Start time • Duration

For the MVP, location search shall be based on the city stored in the professional's address.

FR-12: Search Result Rules

Search results shall only return professionals who:

- Have an approved overall verification status
- Are located in the requested city
- Have free availability slots for the requested time and duration
- Can only view slots available at least 2 hours from now. This gives the professional enough time to accept the booking and plan ahead.

4.5 Booking Management

FR-13: Create Booking

The system shall allow a client to create a booking by selecting:

- Professional
- Address
- Start time
- End time or duration
- Care notes

The system shall validate that the selected professional has the required free availability slots before creating the booking.

FR-14: Prevent Double Booking

The system shall prevent a single availability slot from being linked to more than one booking.

The `booking_slots` table shall be used to reserve the selected hourly slots for a booking.

FR-15: Professional Accept or Reject Booking

The system shall allow the professional to accept or reject a requested booking.

A booking can move from REQUESTED to:

- CONFIRMED
- REJECTED

FR-16: Auto-Reject Unaccepted Booking

The system shall include a scheduled process that automatically rejects bookings that have passed their start time without being confirmed by the professional.

In this case, the system shall:

- Mark the booking as rejected
- Treat the cancellation as caused by the professional for MVP purposes
- Refund the payment record
- Release the reserved slots

FR-17: Cancel Booking

The system shall allow either the client or professional to cancel a booking.

The cancellation policy shall be:

- Full refund if cancelled outside the defined threshold
- No refund if cancelled within the defined threshold

For the MVP, the threshold shall be 48 hours before the booking start time.

FR-18: Mark No-Show

The system shall allow a client to mark a booking as no-show.

FR-19: Complete Booking

The system shall allow a professional to mark a booking as completed after the booking end time.

4.6 Payments

FR-20: Mock Payment on Booking

For the MVP, payment shall be mocked.

When a booking is created, the system shall create a payment record and mark it as paid without integrating with a real payment provider.

FR-21: Payment Record

The system shall create a payment record for each booking.

The payment record shall store:

- Booking reference
- Payment status
- Amount

- Payment provider reference, where applicable
- Refund amount
- Refund timestamp
- Refund reason

FR-22: Refund Handling

The system shall support mocked refund handling by updating the payment status and refund fields.

4.7 Reviews

FR-23: Leave Review

The system shall allow both clients and professionals to leave reviews after a booking.

A review shall contain:

- Booking reference
- Reviewer reference
- Reviewee reference
- Rating
- Comment

Reviews shall only be allowed after a completed booking.

5. Business Rules

Rule ID	Business Rule
BR-1	A professional must not appear in search results unless their overall verification status is approved.
BR-2	A professional is approved only when all required documents are approved and not expired.
BR-3	Availability slots must be created in one-hour increments.
BR-4	Availability slots must start on the hour.
BR-5	A single availability slot must not be booked more than once.
BR-6	Bookings must be created using available slots only.
BR-7	Reviews can only be submitted after a booking is completed.
BR-8	A professional can only complete a booking after the booking end time.
BR-9	Cancellation refund eligibility is based on the 48-hour threshold.
BR-10	Uploaded professional documents are stored in Azure Blob Storage, not directly in the database.

6. Core Data Entities

The core data entities are:

- users
- addresses
- client_profiles
- professional_profiles
- admin_profiles
- document_types
- professional_documents
- availability_slots
- bookings
- booking_slots
- payments
- reviews

These entities and their relationships are defined in the ReadyCare ERD.

7. Non-Functional Requirements

7.1 Security

The system shall protect account and user data through authentication, password hashing, role-based access control, and secure document storage.

7.2 Reliability

The system shall prevent inconsistent booking states, overlapping bookings, and duplicate slot reservations.

7.3 Performance

Search and booking operations should be efficient enough to support the MVP user journey without unnecessary delay.

7.4 Maintainability

The system should separate user, verification, booking, payment, and review logic into clear modules or services.

7.5 Cloud Compatibility

The system should be compatible with Azure deployment, including the use of Azure Blob Storage for professional documents.

8. Out of Scope for MVP

The following are outside the MVP scope:

- Real payment provider integration
- Live chat or messaging
- Mobile application
- Advanced location radius search
- Live push notifications
- Video consultation
- Dynamic pricing
- Automated third-party document verification

9. Summary

ReadyCare's MVP focuses on enabling a complete end-to-end booking workflow between clients and verified care professionals. The core system supports account creation, document verification, professional availability, search, bookings, mocked payments, and reviews.

The ERD supports the main data requirements while keeping the design simple enough for MVP implementation.